Scene2DENM: End-to-End DENM Generation from Traffic Video for Cooperative Autonomous Driving

Kailin Tong*, Baoyun Wang†, Kenan Mujkic*, Christoph Pilz*,
Xingcheng Zhou‡, Selim Solmaz*, Jelena Rubeša-Zrim*, Daniel Watzenig*§,
Arno Eichberger§, Bo Leng†
*Virtual Vehicle Research GmbH, Graz, Austria
†Tongji University, Shanghai, China
‡Technical University of Munich, Munich, Germany
§Graz University of Technology, Graz, Austria

***Abstract*—Cooperative Intelligent Transportation Systems (C-ITS) rely on standardized safety messages for low-latency, interoperable exchange of risk information. Yet existing traffic video understanding benchmarks do not study how visual scene understanding can be translated into protocol-compliant communication, limiting the evaluation of end-to-end autonomous driving systems beyond perception outputs. We introduce Scene2DENM, a new task and benchmark for generating ETSI Decentralized Environmental Notification Messages (DENMs) directly from traffic video. Under explicit schema constraints, the model jointly performs hazard detection, event/cause/sub-cause inference, temporally grounded normalized 2D localization, and evidence-consistent description generation. To support this task, we convert the TUMTraf VideoQA dataset (1001 videos) into a DENM-aligned benchmark comprising 4,316 clips with structured instructions and schema-constrained targets. We then fine-tune TraffiX-Qwen via instruction tuning for structured DENM generation. Evaluation on a held-out validation set of 432 clips shows that fine-tuning removes format violations (parse failures 96–100% $\rightarrow$ 0%) and substantially improves situation detection (accuracy 0.76 $\rightarrow$ 0.96, F1 0.00 $\rightarrow$ 0.91), while achieving grounded localization (IoU up to 0.60) and strong description quality (BLEU-4 0.76, ROUGE-L 0.84, CIDEr 7.94). Scene2DENM provides, to our knowledge, the first quantitative benchmark for structured video-to-standard message generation, bridging traffic scene understanding, standardized V2X communication, and cooperative situational awareness for real-world autonomous driving systems.**

I. Introduction

Connected and Automated Vehicles (CAVs) have achieved significant progress through onboard perception and intelligence. However, single-vehicle sensing remains limited by occlusion, sensing range, and environmental conditions. Vehicle-to-everything (V2X) communication extends situational awareness beyond local perception by enabling safety-relevant information sharing among heterogeneous ITS stations, including onboard units (OBUs) and roadside units (RSUs).

Within the European C-ITS framework, standardized facilities-layer messages support interoperable safety message communication. Cooperative Awareness Messages (CAM) periodically share kinematic state information [1], while Collective Perception Messages (CPM) distribute locally perceived objects to extend environmental awareness [2]. Decentralized Environmental Notification Messages (DENMs) are event-triggered messages generated when hazardous situations are detected, conveying event type and location to relevant ITS stations [3]. Similarly, SAE J2735 defines interoperable V2X message sets in North America [4]. Among these, DENM is particularly suited for communicating abnormal traffic events such as accidents or hazardous road conditions. Other regions, such as China and Japan, have slightly different message encoding systems. But the same principle applies, as demonstrated for DENMs in this paper.

ORCID: Kailin Tong (0000-0002-7040-9237), Kenan Mujkic, (0009-0006-7904-5181), Jelena Rubeša-Zrim (0000-0002-8488-2631), Selim Solmaz (0000-0003-0686-1306), Christoph Pilz (0000-0002-6937-4065), Bo Leng (0000-0003-3513-1708)

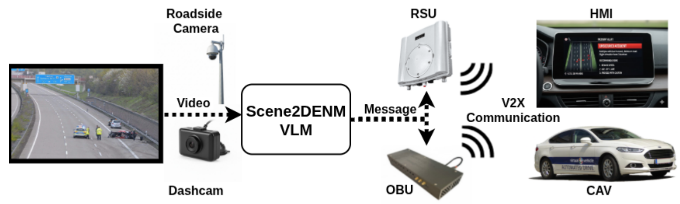

Fig. 1. System overview of **Scene2DENM**. Traffic video is processed by a vision-language model to generate structured DENMs, which are disseminated via RSU/OBU infrastructure to CAVs and human-machine interfaces (HMIs) through V2X communication.

Automatically generating DENMs from visual perception requires more than event detection. Outputs must conform to strict semantic fields and structured schemas to ensure interoperability, reliability, and verifiability. While traffic-scene understanding has advanced through VideoQA and spatio-temporal grounding benchmarks such as TUMTraf VideoQA [5], existing approaches typically produce natural-language answers or localized regions rather than standardized safety messages.

Recent multimodal large language models (MLLMs) integrate visual perception and language reasoning within unified generative frameworks [6]–[9]. However, their outputs are typically unconstrained and not aligned with protocol-defined message schemas, limiting their applicability in safety-critical

V2X systems. Prior efforts have explored integrating large language models with connected and automated vehicles to enable V2X-aware reasoning and standardized communication [10], while multimodal large language models have been investigated for infrastructure-based traffic accident detection [11]. Nevertheless, these approaches often rely on external API-based LLM/VLM services, introducing additional communication overhead and raising data security concerns, thereby hindering deployment in practical V2X environments.

Figure 1 illustrates the proposed Scene2DENM framework. Traffic video captured by roadside cameras is processed by a vision-language model to generate ETSI-compliant DENM messages, which are then disseminated through RSU/OBU infrastructure to connected and automated vehicles (CAVs) and driver human–machine interfaces (HMIs) via V2X communication.

By coupling multimodal perception with schema-constrained message generation, Scene2DENM establishes an end-to-end pipeline from visual scene understanding to standardized cooperative safety messaging. As demonstrated in [10], such V2X messages can extend the situational awareness of autonomous vehicles beyond their onboard sensing range, enabling proactive maneuvers such as lane changes or trajectory adjustments to avoid hazards (e.g., road damage). This cooperative information sharing can improve both the safety and ride comfort of autonomous driving systems.

We present Scene2DENM as an initial step toward protocol-aware video understanding for cooperative driving systems. The main contributions of this work are summarized as follows:

- We formulate a new research problem for end-to-end generation of ETSI-compliant DENMs from traffic video, unifying situation detection, event semantics, spatio-temporal grounding, and structured message generation in a single benchmark setting.
- We construct a DENM-aligned benchmark from TUM-Traf VideoQA [5] by converting roadside traffic videos into instruction-following samples with schema-constrained JSON targets and grounded temporal and spatial annotations, and we develop and evaluate a supervised fine-tuning framework based on TraffiX-Qwen [5] together with an evaluation protocol covering format compliance, situation detection, message correctness, localization, and description quality.
- We further present an initial hardware-oriented feasibility test showing that generated structured outputs can be integrated into an embedded communication pipeline without manual reformulation.

II. Methodology

A. Problem Formulation

We cast DENM generation as a structured multimodal reasoning problem. Given a traffic video clip and a task prompt, the model must produce an ETSI-compliant DENM message in JSON format. The output comprises: (i) binary situation detection, (ii) event and cause-code classification according to the DENM standard [3], (iii) spatiotemporal localization via two temporal anchors with normalized 2D bounding boxes, and (iv) a concise natural-language description grounded in the scene. The task, therefore, integrates classification, localization, and schema-constrained generation within a unified framework.

B. Dataset Construction

We derive a DENM-oriented dataset from the TUMTraf VideoQA benchmark [5]. Raw traffic videos are segmented into **5-second clips**, a duration optimized for roadside infrastructure cameras and Low-Latency V2X communication applications. Each clip is transformed into an instruction-following instance pairing video input with a schema-constrained DENM JSON target.

Our annotation strategy follows several rigorous rules: (i) **Situation Marking**: For the earliest occurring event, we always mark the first and last frames of the situation duration. In cases of simultaneous events, only the most critical situation with the highest impact is annotated. (ii) **Spatial Grounding**: Normalized spatiotemporal bounding boxes include all objects relevant to the primary situation while strictly excluding unrelated entities. For large-scale environmental hazards (e.g., fog), the entire image frame serves as the bounding box. (iii) **Protocol Compliance**: All fields are strictly aligned with the **ETSI TS 103 831** (DENM) standard. We adopt a conservative reporting threshold in our labels to prevent over-sensitivity in safety-critical automated responses.

To address the inherent class imbalance in traffic situation detection, we follow a strategic sampling distribution during training [12]. We maintain a ratio of **25% situation events** to **75% non-event samples**. Within the non-event category, we deliberately avoid a preponderance of "easy negatives" (e.g., clear roads under ideal conditions). This distribution forces the model to learn fine-grained discriminative features rather than relying on global scene statistics.

C. Model Architecture

We fine-tune TraffiX-Qwen at two parameter sizes (0.5B and 7B) in an instruction-following setting conditioned on visual tokens and structured metadata. As shown in Figure 2, the model adopts a SigLIP vision encoder paired with a Qwen2 language backbone. Video frames are encoded into visual tokens, projected into the language embedding space, and jointly processed with tokenized instruction prompts to generate a structured DENM JSON response.

We reuse the pre-trained TraffiX-Qwen vision encoder and projector weights and keep them frozen during training to preserve their learned visual representations and reduce computational cost. The Qwen2 language backbone is fine-tuned either with full-parameter supervised fine-tuning for maximum adaptability or with LoRA (Low-Rank Adaptation) to achieve parameter-efficient training with lower memory and compute requirements. We perform supervised fine-tuning at

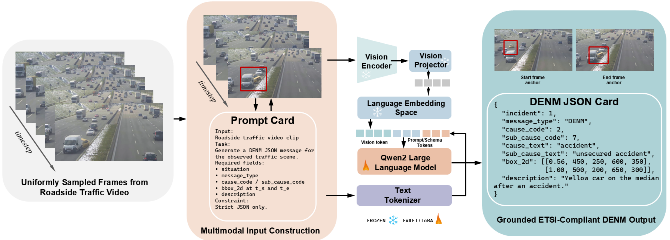

Fig. 2. Scene2DENM architecture for schema-constrained video-to-DENM generation, adapted from [5].

101 frames per video to capture sufficient temporal context for accurate scene understanding and downstream safety message communication.

D. Fine-tuning Strategy

All models are trained using supervised fine-tuning (SFT) for one epoch on 3,884 DENM QA training samples. We reuse the pre-trained TraffiX-Qwen vision encoder and projector weights and keep them frozen to preserve learned visual representations and reduce computational cost.

We evaluate three fine-tuned variants: **0.5B Full FT**, **0.5B LoRA**, and **7B LoRA**, alongside pre-trained base models (0.5B, 7B) as zero-shot references. All configurations use 101 input frames to maximize temporal context for situation analysis.

For the **0.5B model**, we compare (i) full-parameter fine-tuning (no pooling) and (ii) LoRA with stride-2 spatial pooling. For the **7B model**, we apply LoRA with stride-4 spatial pooling, reducing the 729 visual tokens per frame to approximately 49 tokens, enabling efficient training on four NVIDIA A100 GPUs (80 GB VRAM each).

Training employs a mixture of message-generation and evidence-selection prompts to promote structured reasoning and grounded outputs, while monitoring hallucination and over-assertion.

E. Evaluation Protocol

Evaluation is conducted on a held-out TUMTraf DENM QA validation set comprising 432 samples (104 positive situations and 328 negatives). Outputs are first validated using a strict DENM JSON schema checker to ensure structural compliance.

We report the following metrics:

- **Format compliance:** JSON parse failures.
- **Situation detection:** accuracy, precision, recall, and F1.
- **Message correctness:** message-type and cause-code+text accuracy on true-positive (TP) situations only.
- **Localization:** temporal L_1, box-center L_2, and IoU (computed on TP cases).
- **Description quality:** BLEU-4, ROUGE-L, and CIDEr.

III. Results: Initial Observations

A. Key Findings

a) Evaluated Configurations.: We compare pre-trained base models (0.5B, 7B) with three fine-tuned variants (0.5B Full FT, 0.5B LoRA, 7B LoRA). Base models serve as zero-shot baselines, while fine-tuned models follow the strategy described in Section II-D.

TABLE I

Comprehensive DENM evaluation results on the 432-sample validation set. Best performing models for each metric are highlighted in **bold**.

Metric	0.5B Parameters			7B Parameters	
	Base	FT (Full)	LoRA	Base	LoRA
Parse Failures ↓	432	**0**	**0**	432	**0**
Situation TP ↑	0	**95**	88	0	76
Situation TN ↑	**328**	318	320	**328**	319
Situation FP ↓	**0**	10	8	**0**	9
Situation FN ↓	104	**9**	16	104	28
Accuracy ↑	0.76	**0.96**	0.94	0.76	0.91
Precision ↑	0.00	0.91	**0.92**	0.00	0.89
Recall ↑	0.00	**0.91**	0.85	0.00	0.73
F1-Score ↑	0.00	**0.91**	0.88	0.00	0.80
Msg. Type Acc. ↑*	–	**1.00**	**1.00**	–	**1.00**
Code+Text Acc. ↑*	–	0.83	0.89	–	**0.93**
Temporal L_1 Error ↓*	–	**0.00**	**0.00**	–	**0.00**
Box Center L_2 ↓*	–	**123.57**	218.03	–	244.38
Box IoU ↑*	–	**0.60**	0.40	–	0.45
BLEU-4 ↑	0.00	**0.76**	0.73	0.00	0.73
ROUGE-L ↑	0.00	**0.84**	0.82	0.00	0.82
CIDEr ↑	0.00	**7.94**	7.73	0.00	7.74

*Evaluated on true positive (TP) situation detections only.

Table I highlights two main factors influencing performance: model scale (0.5B vs. 7B) and adaptation strategy (full vs. LoRA).

b) Format & Detection.: Base models and the 7B LoRA fail entirely (100% parse failures). In contrast, both 0.5B fine-tuned variants and the 7B LoRA variant achieve zero failures. The 0.5B Full FT model achieves the highest accuracy (0.96) and F1 (0.91), whereas the 7B LoRA variant achieves a competitive F1 of 0.80 and an accuracy of 0.91.

c) Grounded Reasoning.: Spatial localization reveals a strong advantage for the 0.5B Full FT model (IoU 0.60, $L_2 = 123.57$), significantly outperforming the LoRA variants. Interestingly, the **7B LoRA model achieves the highest Code+Text accuracy (0.93)**, outperforming both 0.5B variants (0.83–0.89). This suggests that the larger 7B pre-trained backbone excels at structured field generation once the instruction-following mapping is learned via LoRA, despite being slightly less spatially precise.

d) Language Quality.: Textual metrics peak with the 0.5B Full FT model (CIDEr 7.94), confirmation that smaller backbones with total weight updates can achieve high fidelity. However, both the 0.5B and 7B LoRA models maintain strong performance on BLEU-4 and CIDEr, confirming the robustness of schema-constrained generation across scales.

B. Hardware Test

To complement the offline quantitative evaluation, we conducted a hardware-oriented feasibility test using a communication setup with two On-Board Units (OBUs), as shown in Figure 3. The objective of this test was to verify that the language model's structured outputs can be directly mapped to standardized C-ITS messages and delivered end-to-end through an embedded V2X communication pipeline without manual intervention.

a) Experimental Setup: The test environment consisted of two Unex SOM-301E OBUs, based on the *VehicleCAPTAIN toolbox* developed by Pilz et al. [13], connected to separate development workstations. Both workstations ran ROS 2 Humble within Docker environments, while communication between the workstations and the OBUs was established via a Zenoh DDS bridge operating in client mode. The two OBU devices communicated with each other over an ITS-G5 (IEEE 802.11p) radio interface. To ensure consistent timing measurements, both the OBU devices and the Docker environments were synchronized using the same NTP server (time.tugraz.at).

b) Message Generation and Transmission Pipeline: On the sender side, a Python script reads the JSON output of the language model, which includes key semantic fields such as *cause_code*, *sub_cause_code*, and *description*, and maps them to a standardized DENM structure based on ETSI EN 302 637-3 [3]. The mapping and message construction leverage the open-source ROS 2 implementation of ETSI ITS messages provided by Küppers et al. [14]. The required message containers are populated, including management information (e.g., event position and timestamp) and situation information (cause and sub-cause codes). The resulting message is then encoded into ASN.1 UPER binary format and converted into a UDP packet compliant with the ETSI ITS protocol stack. The packet is subsequently transmitted via the OBU over the ITS-G5 (IEEE 802.11p) radio interface.

c) Message Reception: On the receiving side, the second OBU receives the ITS-G5 radio frame and forwards the decoded content to the ROS 2 environment using the same ETSI ITS ROS 2 implementation [14]. Successful reception and the correctness of the DENM message are verified by inspecting the decoded data.

d) Results and Latency Analysis: The experiment confirms that the complete pipeline—from structured JSON output to standards-compliant radio transmission—operates reliably without requiring any manual data transformation. To quantify system performance, latency measurements were conducted using three timestamps: T_1 (script start time), T_2 (message transmission from the first OBU), and T_3 (message reception at the second OBU). The measured delays are $T_2 - T_1 \approx 2.6\,\mathrm{ms}$ (processing and message generation), $T_3 - T_2 \approx 23.3\,\mathrm{ms}$ (wireless transmission over ITS-G5), and $T_3 - T_1 \approx 25.9\,\mathrm{ms}$ (end-to-end latency). These results indicate that the majority of the latency is attributable to wireless transmission, while the processing overhead remains minimal. The overall end-to-end latency of approximately $26\,\mathrm{ms}$ demonstrates that the system satisfies the requirements of time-critical C-ITS applications.

Fig. 3. The Vehicle CAPTAIN hardware, designed and built by Virtual Vehicle Research. It can be used as a development board, On-Board Unit (OBU), or Roadside Unit (RSU).

e) Initial Conclusion: The results confirm the technical feasibility of integrating VLM-based hazard detection with real-world V2X systems. Structured outputs can be seamlessly converted into standardized DENM messages and reliably processed in a real-world communication setup, thereby supporting their applicability in cooperative autonomous driving. Compared to the VLM inference time (typically 3–5 seconds), V2X communication latency is negligible. Therefore, future work should primarily focus on reducing inference time to enable real-time deployment.

IV. Conclusion and Future Work

This exploratory study suggests a hierarchy of influence: (i) supervised fine-tuning is essential for structured DENM generation; (ii) full-parameter adaptation primarily improves spatial localization; and (iii) increasing temporal context (i.e., more frames) does not compensate for limited weight adaptation.

Furthermore, the study confirms the technical feasibility of integrating VLM-based hazard detection with real-world V2X systems, where structured outputs can be directly transformed into standardized DENM messages and reliably processed within a real communication setup.

Several limitations remain. First, although we expanded the evaluation to 432 samples (104 positives), the sample size for certain rare incident types remains limited.

Moreover, uniformly increasing the number of sampled frames does not necessarily improve performance. Many frames within a single scenario are redundant, particularly in largely static situations (e.g., reduced visibility due to fog), where a single representative frame may suffice. In contrast, dynamically evolving events (e.g., unsecured accidents) require temporally focused sampling around critical transitions. The current uniform sampling strategy, therefore, introduces unnecessary computational overhead and should be replaced by a more adaptive, context-aware mechanism.

Future work will focus on four particularly promising directions: (1) training on more general *fixed-duration video snippets* (e.g., 5s) collected across more diverse locations, traffic conditions, and countries to improve model robustness and generalization; (2) replacing uniform sampling with *key-frame extraction* or dynamic sampling strategies (e.g., temporal difference analysis or attention-based selection); (3) enlarging and diversifying the dataset to mitigate potential performance ceilings; and (4) validating the system on on-board or roadside units using live video streams, with a focus on reducing VLM inference latency to enable real-time deployment.

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